

Semigroups and Groups

Semigroup and Groups

9.1 二元运算/Binary Operations Revisited

9.2 半群/Semigroups

9.3 乘积半群与商半群/Products and Quotients of Semigroups

9.4 群Groups

9.5 乘积群与商群Products and Quotients of Groups

9.4 Groups – 群

Groups

Definition (定义)

Abelian (阿贝尔群)

Properties of groups (群性质)

Finite groups (有限群)

Subgroups (子群)

Group 群定义

A group $(G, *)$ is a monoid, with identity e , that has the additional property that for every element $a \in G$ there exists an element $a' \in G$, such that

$$a * a' = a' * a = e.$$

group

A **group** $(G, *)$ is a set together with a binary operation $*$ on G such that, for any elements a, b , and c in G . $\forall a, b, c \in G$

➤ $a * b \in G$ -- 封闭

➤ $(a * b) * c = a * (b * c)$ -- 结合

a unique element e in G such that

➤ $a * e = e * a$ -- 单位元

an element $a' \in G$, called an inverse of a and written as a^{-1} , such that

➤ $a * a' = a' * a = e$ or

$aa' = a'a = e$ or $aa^{-1} = a^{-1}a = e$ ----- 每个元素有逆元

群的常用术语

1. 平凡群:只含单位元的群 $\{e\}$
2. 交换群:Abel群
3. 有限群与无限群
4. 群 G 的阶, G 的基数,通常有限群记为 $|G|$
5. 元素 a 的 n 次幂

$$a^n = \begin{cases} e & n = 0 \\ a^{n-1}a & n > 0 \\ (a^{-1})^n & n < 0 \end{cases}$$

- 元素 a 的阶 $|a|$,使得 $a^k = e$ 成立的最小正整数 k
注:有限群的元素都是有限阶,是群的阶的因子,
反之,元素都是有限阶的群不一定是有限群

Abelian – 阿贝尔群

- A group G is said to be **Abelian** if $ab = ba$ for all elements a and b in G .

Table 9.2

	e	a
e	e	a
a	a	e

Table 9.4

	e	a	b
e	e	a	b
a	a	b	e
b	b	e	a

Table 9.5

	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	e	a
c	c	b	a	e

阿贝尔群的乘法表是关于主对角线对称的

Example 4

Let

G be the set of all nonzero real numbers and
 $a * b = ab/2$.

Show

$(G, *)$ is an Abelian group.

Example 4 - Solution

* is a **binary operation** —运算封闭

If a and b are elements of G ,
then $ab/2$ is a nonzero real number and
hence is in G .

associativity ---可结合

$$(a * b) * c = (ab/2) * c = (ab)c/4$$

$$a * (b * c) = a * (bc/2) = a(bc)/4 = (ab)c/4$$

* is associative.

Example 4 - Solution

2 is the identity (有单位元)

$$a * 2 = (a)(2)/2 = a = (2)(a)/2 = 2 * a.$$

$a' = 4/a$ is an inverse of a --- 逆元

$$\begin{aligned} a * a' &= a * 4/a = a(4/a)/2 = 2 = (4/a)(a)/2 \\ &= (4/a) * a = a' * a. \end{aligned}$$

Abelian

$$a * b = ab/2 = ba/2 = b * a$$

So, G is an Abelian group.

Theorem 1

逆元唯一

Let G be a group. Each element a in G has only one inverse in G .

Proof : 反证法

Let

a' and a'' be inverses of a .

Then

$$a' = a'e = a'(aa'') = (a'a)a'' = ea'' = a''.$$

Theorem 2-消去律

Let

G be a group and

a , b , and c be elements of G .

Then

left cancellation – 左消去律

$ab = ac$ implies that $b = c$

right cancellation – 右消去律

$ba = ca$ implies that $b = c$

Theorem 2 - Proof

Left cancellation

Suppose that $ab = ac$.

$$a^{-1}(ab) = a^{-1}(ac)$$

$$(a^{-1}a)b = (a^{-1}a)c \text{ by associativity}$$

$$eb = ec \quad \text{by the definition of an inverse}$$

$$b = c \quad \text{by definition of an identity.}$$

Right cancellation

The proof is similar to above.

Theorem 3

Let

G be a group and

a and b be elements of G .

Then

$$(a^{-1})^{-1} = a.$$

$$(ab)^{-1} = b^{-1}a^{-1}$$

Theorem 3 - Proof

➤ $\Rightarrow (a^{-1})^{-1} = a$

$$\begin{cases} (a^{-1})^{-1}(a^{-1}) = e \\ aa^{-1} = e \end{cases} \Rightarrow (a^{-1})^{-1}(a^{-1}) = aa^{-1}$$

the inverse of an element is unique,

So, $(a^{-1})^{-1} = a$

➤ $\Rightarrow (ab)^{-1} = b^{-1}a^{-1}$

$$(ab)(b^{-1}a^{-1}) = a(b(b^{-1}a^{-1})) = a((bb^{-1})a^{-1}) = a(ea^{-1}) = aa^{-1} = e$$

similarly, $(b^{-1}a^{-1})(ab) = e$

so $(ab)^{-1} = b^{-1}a^{-1}$ (注：同时证左逆右逆都成立)

Theorem 4

Let

G be a group, and
 a and b be elements of G

群方程有唯一解

Then

- The equation $ax = b$ has a unique solution in G .
- The equation $ya = b$ has a unique solution in G .

证明

(a) 令 $x = a^{-1}b$ 代入方程, $\Rightarrow a(a^{-1}b) = (a a^{-1})b = eb = b$, 所以元素 $x = a^{-1}b$ 是方程 $ax = b$ 的一个解。

现在假设 x_1 和 x_2 是方程 $ax = b$ 的两个解,

那么 $ax_1 = b$, $ax_2 = b$ 因此 $ax_1 = ax_2$

由定理2推出 $x_1 = x_2$

(b) 证明类似 (a)

定理4-续

设 G 是半群, 如果 $\forall a, b \in G$, 方程 $ax=b$ 和 $ya=b$ 在 G 中有解, 则 G 为群。

证明 (只要找单位元, 和逆元即可)

$\forall b \in G$, 方程 $bx=b$ 的解 记为 e

$\forall a \in G$, 方程 $yb=a$ 的解记为 c , 即 $cb=a$

$$ae = (cb) e = c (be) = cb = a$$

所以 e 为右单位元

因为 a 是任意的, 方程 $ax=e$ 有解, 所以 a 有右逆元。

G 有单位元, 每个元素有逆, 所以 G 是群

Finite group – 有限群

If G is a group that has a finite number of elements, G is said to be a *finite group*, and the *order*(阶) of G is the number of elements $|G|$ in G .

A finite group can be represented in the form of the multiplication table.

$$ax = b$$

$$ya = b$$

*	a1	a2	...	an
a1				
a2				
...			$a_i * a_j$	
an				

群中任一元素 a ，在每行（列）仅且仅出现一次

Group of order 1, 2

If G is a group of order 1, then

$$G = \{e\}, \text{ and } ee = e.$$

Let $G = \{e, a\}$ be a group of order 2.

The blank can be filled in by e or by a ?

Table 9.1

	e	a
e	e	a
a	a	?



Table 9.2

	e	a
e	e	a
a	a	e

Nonisomorphic groups of order 3

Let $G = \{e, a, b\}$ be a group of order 3.

Table 9.3

	e	a	b
e	e	a	b
a	a		
b	b		



Table 9.4

	e	a	b
e	e	a	b
a	a	b	e
b	b	e	a

Groups of order 4—4阶群

Let $G = \{e, a, b, c\}$ be a group of order 4

构造更多的运算表？

Table 9.5

	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	e	a
c	c	b	a	e

Table 9.6

	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	a	e
c	c	b	e	a

Table 9.7

	e	a	b	c
e	e	a	b	c
a	a	b	c	e
b	b	c	e	a
c	c	e	a	b

Table 9.8

	e	a	b	c
e	e	a	b	c
a	a	c	e	b
b	b	e	c	a
c	c	b	a	e

群的性质小结

幂运算规则:

1. $(a^{-1})^{-1} = a$

2. $(ab)^{-1} = b^{-1}a^{-1}$ 可以推广到 n 个元素的积的逆。

3. $(a^n)^{-1} = (a^{-1})^n$

4. $a^m a^n = a^{m+n}$

5. $(a^m)^n = a^{mn}$

6. 若 G 为Abel群, 则 $(ab)^n = a^n b^n$

群的性质

6. 群中除了单位元 e 外，没有其他的幂等元。
（假设 $a*a=a$ ，因为 $a*e=a$ ，根据消去律 $a=e$ ）

消去律：

$$1. \quad ab=ac \Rightarrow b=c \quad \text{--左消去}$$

$$2. \quad ba=ca \Rightarrow b=c \quad \text{--右消去}$$

Example 5

Let $B = \{0, 1\}$, and let $+$ be the operation defined on B as follows:

$+$	0	1
0	0	1
1	1	0

Then B is a group.

2阶群每个元素是它自身的逆

An important group

Given the equilateral triangle with vertices 1, 2, and 3

Consider its symmetries.

等边三角形顶点的置换

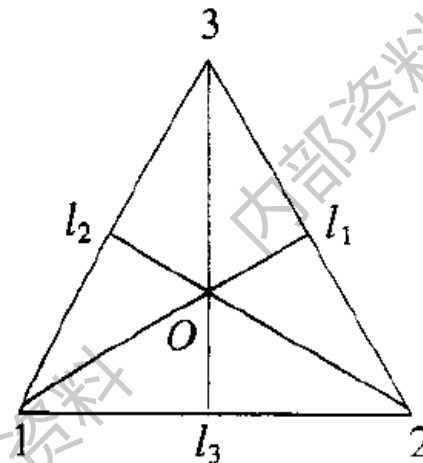


Figure 9.3

Symmetries of the triangle

There are counter-clockwise rotations f_2 , f_3 , f_1 of the triangle about O through 120° , 240° , 360° (or 0°) respectively.

f_1 , f_2 , f_3 can be written as the permutations.

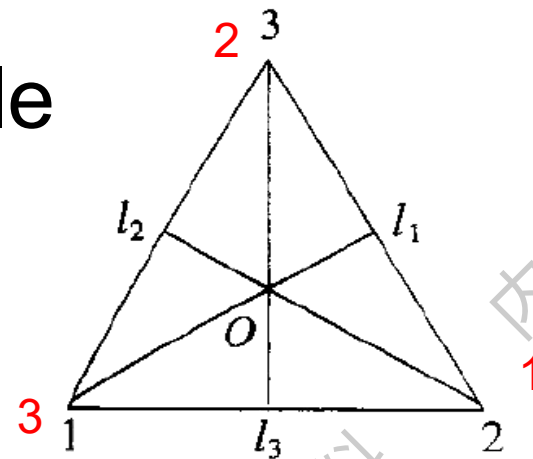


Figure 9.3

f_1 , f_2 , f_3 分别代表绕 O 点逆时针旋转 360° , 120° , 240°

Symmetries of the triangle

Three additional symmetries of the triangle are g_1 , g_2 , and g_3 , by reflecting about the lines l_1 , l_2 , and l_3 , respectively.

Denote these reflections as the following permutations:

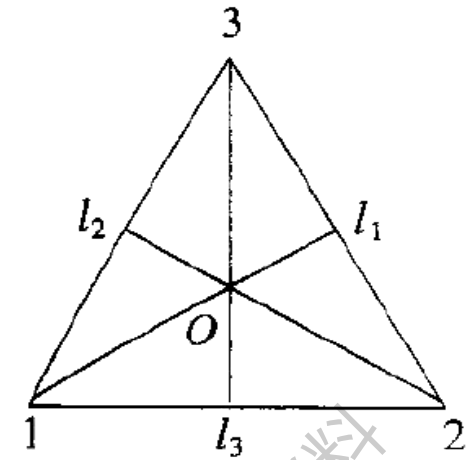


Figure 9.3

Group of symmetries of the triangle 三角形对称群

Let $S_3 = \{f_1, f_2, f_3, g_1, g_2, g_3\}$ and introduce the operation $*$, *followed by*, on the set S_3

*表示合成运算，如 $f_2 * g_2$ 表示，先进行 g_2 变换，再进行 f_2 变换

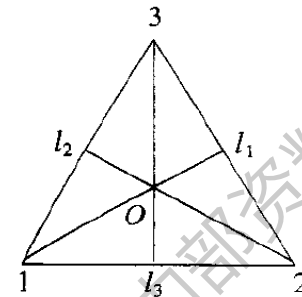
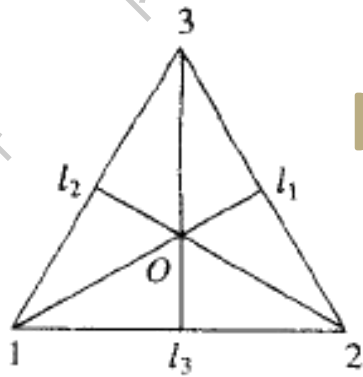
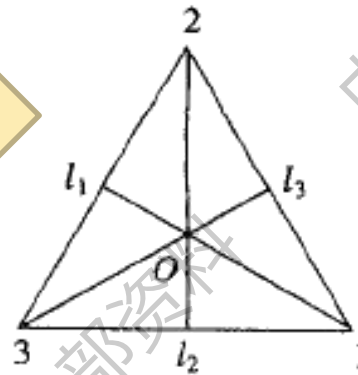


Figure 9.3

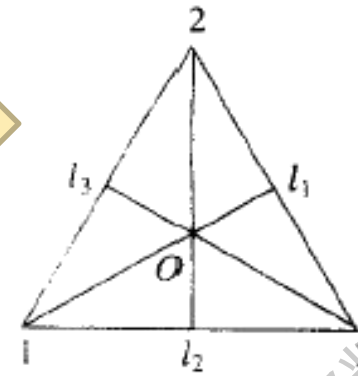
Compute $f_2 \circ g_2$ geometrically



Given triangle



Triangle resulting after applying f_2



Triangle resulting after applying g_2 to the triangle at the left

Compute $f_2 \circ g_2$ algebraically

To compute $f_2 \circ g_2$ algebraically, we compute $f_2 \circ g_2$.

$$\text{So } f_2 \circ g_2 = g_1.$$

Example 7

The set of all permutations of n elements is a group of order $n!$ under the operation of composition.

This group is called the *symmetric group on n letters* (n 个字母上的对称群) and is denoted by S_n .

由部分置换构成的群称为置换群(permutation group)

$$S_4 \rightarrow 4! \rightarrow \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix}, \quad \dots$$

Subgroup – 子群

Let H be a subset of a group G such that

- (a) The identity e of G belongs to H .
- (b) If a and b belong to H , then $ab \in H$.
- (c) If $a \in H$, then $a^{-1} \in H$.

Then H is called a **subgroup** of G .

判断子群的方法一: $H \subseteq G$

➤ $e \in H$

➤ If $\forall a, b \in H$, then $ab \in H$ —— 封闭

➤ If $a \in H$, then $a^{-1} \in H$ —— 都有逆

Subgroup – 子群

判断子群的方法二: 非空 $H \subseteq G$

➤ $\forall a, b \in H, a b^{-1} \in H$

含单位元

证: 1. H 非空, $\exists b \in H$

2. $\exists b \in H \Rightarrow b b^{-1} \in H \Rightarrow e \in H$ (根据 $a b^{-1} \in H$)

3. $\forall a \in H \Rightarrow e a^{-1} \in H \Rightarrow a^{-1} \in H$

4. $\forall a, b \in H \Rightarrow a^{-1} \in H$ and $b^{-1} \in H$

证都有逆元

$ab^{-1} \in H \Rightarrow a(b^{-1})^{-1} \in H$

$\Rightarrow ab \in H$

证封闭

Examples9

Let G be a group. Then G and $H = \{e\}$ are subgroups of G , called the *trivial subgroups* (平凡子群) of G .

$H = \{e\}$ 是 G 的平凡子群

Examples10

Consider S_3 , the group of symmetries of the equilateral triangle. It is easy to verify that $H = \{f_1, f_2, f_3\}$ is a subgroup of S_3 .



Example 11

Let A_n be the set of all even permutations in the group S_n . It can be shown from the definition of even permutation that A_n is a subgroup of S_n called the *alternating group on n letters*. (n 次变错群)

生成子群的一种方法： 元素 a 的幂生成的子群

Let

G be a group and $a \in G$.

Define

a^n as $aa\ldots a$ (n factors), for $n \in \mathbb{N}$ and

a^0 as e .

a^{-n} as $a^{-1}a^{-1}\ldots a^{-1}$ (n factors).

if n and m are any integers, then $a^n a^m = a^{n+m}$.

It is easy to show that

$H = \{ a^i \mid i \in \mathbb{Z} \}$ is a subgroup of G . (generated by a)

1. 非空

2. 封闭

3. 有单位元, $a^0 = e = aa^{-1} \in H$

4. 存在逆元, $a^i = a^i a^{-i} = e$, $a^{-i} \in H$

5. 可结合

元素 a 的幂生成的子群 (续)

例： $(\mathbb{Z}, +)$, 单位元0

元素 $2 \in \mathbb{Z}$, 生成的群?

$$2^0 = 0, 2^1 = 2, 2^2 = 4, 2^3 = 6, \dots$$
$$2^{-1} = -2, \dots$$

2 可生成偶数群 $(T, +)$ 是 $\mathbb{Z}$ 的子群

由1可生成子群吗?

思考题：

- 群 G 的有限非空子集 $H \subseteq G$
- 如果运算在 H 上是封闭的， H 是否是 G 的子群？
- S3 子群的个数

Cyclic group – 循环群

Recall the subgroup generated by a . Now
Check Table 9.7



Table 9.5

	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	e	a
c	c	b	a	e

Table 9.6

	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	a	e
c	c	b	e	a

Table 9.7

	e	a	b	c
e	e	a	b	c
a	a	b	c	e
b	b	c	e	a
c	c	e	a	b

Table 9.8

	e	a	b	c
e	e	a	b	c
a	a	e	e	b
b	b	e	c	a
c	c	b	a	e

Such group is called *cyclic group*.

---生成元 a 生成 G 的所有元素

循环群至少含一个生成元

循环群-续

定义: $G = \{ a^k \mid k \in \mathbb{Z} \}$, $a \in G$,
称 G 为循环群, a 为 G 的生成元。

分类:

- 生成元的阶无限, 则 G 为无限循环群。生成元为 a 和 a^{-1}
- 生成元 a 为 n 阶元, 则 G 为 n 阶循环群, 生成元的个数为 $\psi(n)$ —数论中的欧拉函数, 表示与 n 互质的数的个数
- 若 G 为无限循环群, 生成子群, 除 $[e]$ 外, 都是无限子群, 生成元为 a 和 a^{-1} 。
- 若 G 为 n 阶有限循环群, 则生成子群的阶是 n 的正因子。对于每个正因子 d , G 有且只有一个 d 阶子群。

循环群—续

实例:

$(\mathbb{Z}, +)$ 为无限循环群? 生成元1和-1

$(\mathbb{Z}_n, \oplus)$, 为n阶循环群

$(\mathbb{Z}_6, \oplus)$, [1]和[5] 是生成元;

生成子群阶分别为1,2,3,6

由[0]生成1阶子群 {[0]}

由[1]生成6阶子群 $\mathbb{Z}_6$

由[2]生成3阶子群 {[0],[2],[4]}

由[3]生成2阶子群 {[0],[3]}

Theorem 5—群同态的特性

Let

$(G, *)$ and $(G', *')$ be two groups, and
 $f: G \rightarrow G'$ be a homomorphism from G to G' .

Then

- (a) If e is the identity in G and e' is the identity in G' , then $f(e) = e'$.
- (b) If $a \in G$, then $f(a^{-1}) = (f(a))^{-1}$.
- (c) If H is a subgroup of G , then $f(H) = \{f(h) \mid h \in H\}$ is a subgroup of G' .

$$H \subseteq G \Rightarrow f(H) \subseteq G':$$

$$\begin{aligned} f: G \rightarrow G' &\Rightarrow f(a * a^{-1}) = f(a) *' f(a^{-1}) \\ &\Rightarrow f(e) = f(a) *' f(a^{-1}) \Rightarrow f(a) = f(a^{-1})^{-1} \end{aligned}$$

Proof - (a)

Let

$$x = f(e)$$

Then

$$x *' x = f(e) *' f(e) = f(e * e) = f(e) = x$$

so $x *' x = x$ *Multiplying both sides by x^{-1} on the right*

$$x *' x *' x^{-1} = x *' x^{-1} = e'$$

Thus $f(e) = e'$

Proof - (b)

Let

$$a \in G$$

Then

$$a * a^{-1} = e$$

$$\text{so } f(a * a^{-1}) = f(e) = e'$$

$$f(a) *' f(a^{-1}) = e'$$

$$\text{Similarly, } f(a^{-1}) *' f(a) = e'$$

Hence

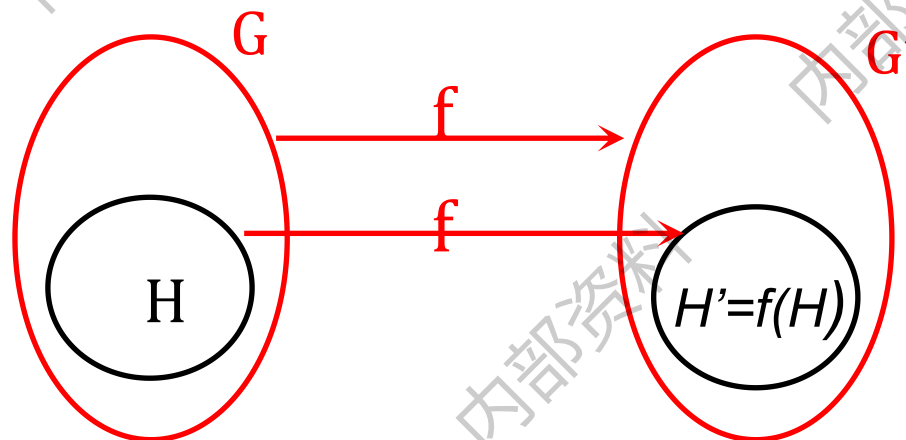
$$f(a^{-1}) = (f(a))^{-1}$$

Proof - (c) 1

解读(c)

f 是 $G \rightarrow G'$ 上的同态，那么对于 G 的一个子群 H ，经过同样的 f 操作后得到一个 H' ($H \rightarrow H'$)，那么 H' 也是 G' 的一个子群

$$H' = f(H) = \{f(h) | h \in H\}$$



Proof - (c) 2

证明 H' 是一个子群:

Step1(证明 G' 的单位元 $f(e)$ 也在 $f(H)$ 内):

$\because H$ 是 G 的子群, e 是 G 的单位元

$\therefore e \in H$

$\because f(H) = \{f(h) | h \in H\}$

$\therefore f(e) \in f(H)$

Proof - (c) 3

任取 $f(H)$ 中的两个元素 $f(h1)$ 和 $f(h2)$, 这里 $h1, h2 \in H$ 。

Step2(封闭性):

$f(h1) * f(h2) = f(h1 * h2)$ —— 利用 $f: G \rightarrow G'$ 同态性质

$\because h1, h2$ 在 H 内满足运算封闭性

$\therefore$ 存在 $h1 * h2 = h3 \in H$

$\therefore f(h1) * f(h2) = f(h3) \in f(H)$

封闭性证毕.

Proof - (c) 4

Step3(逆元存在):

对于任意的 $f(h) \in f(H)$

$\because h \in H$

$\therefore$ 存在 h 的逆元, $h^{-1} \in H$,

$$\therefore f(h^{-1}) * f(h) = f(h^{-1} * h) = f(e) = f(h * h^{-1}) = f(h) * f(h^{-1})$$

逆元存在证毕.

Note

Remember that an isomorphism preserves all properties defined in terms of the group operations.

同构应该保持群运算所定义的所有性质

Example 16

群 S_3 和 Z_6 都是6阶群，但 S_3 不是Abelian
而 Z_6 是Abelian
-----它们不同构

Example 17

- The groups with multiplication Tables 9.6, 9.7, and 9.8 are isomorphic.

Table 9.5

	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	e	a
c	c	b	a	e

Table 9.6

	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	a	e
c	c	b	e	a

- The group with multiplication Table 9.5 is called the *Klein 4 group* (克莱因四元素群) and it is denoted by V .

Table 9.7

	e	a	b	c
e	e	a	b	c
a	a	b	c	e
b	b	c	e	a
c	c	e	a	b

Table 9.8

	e	a	b	c
e	e	a	b	c
a	a	c	e	b
b	b	e	c	a
c	c	b	a	e

- Tables 9.5 and 9.6 不同构?

- Observe that if x is any element in the group determined by Table 9.5, then $x^2 = e$

Z_4 的子群?

例： $(G, *)$ 是一个群， $a \in G$ ，
 $\forall x \in G, f(x) = a x a^{-1}$
证 f 是 G 上的自同构

Homework

12, 18, 28, 34 @9.4, 教材第6版

例题 1 设 $A = \{a, b, c, d\}$, 在 A 上定义一个二元运算, 如表 5-8.2 所示。又设 $B = \{\alpha, \beta, \gamma, \delta\}$, 在 B 上定义一个二元运算, 如表 5-8.3 所示。证明 $\langle A, \star \rangle$ 和 $\langle B, * \rangle$ 是同构的。

表 5-8.2

$\star$	a	b	c	d
a	a	b	c	d
b	b	a	a	c
c	b	d	d	c
d	a	b	c	d

表 5-8.3

$*$	α	β	γ	δ
α	α	β	γ	δ
β	β	α	α	γ
γ	β	δ	δ	γ
δ	α	β	γ	δ

证明 考察映射 f , 使得

$$f(a) = \alpha, f(b) = \beta$$

$$f(c) = \gamma, f(d) = \delta$$

显然, f 是一个从 A 到 B 的双射, 由表 5-8.2 和表 5-8.3 容易验证 f 是由 $\langle A, \star \rangle$ 到 $\langle B, * \rangle$ 的一个同态。因此, $\langle A, \star \rangle$ 和 $\langle B, * \rangle$ 是同构的。

如果考察映射 g , 使得

$$g(a) = \delta, g(b) = \gamma$$

$$g(c) = \beta, g(d) = \alpha$$

那么, g 也是由 $\langle A, \star \rangle$ 到 $\langle B, * \rangle$ 的一个同构。

当两个代数系统是同构的话, 它们之间的同构映射可以是不唯一的。

例 8 如表 5-8.8 所确定的两个代数系统 $\langle A, \star \rangle$ 和 $\langle B, * \rangle$ 。

表 5-8.8

$\star$	α	β	γ	δ	ϵ	ζ
α	α	β	α	α	γ	δ
β	β	α	γ	β	γ	ϵ
γ	α	γ	α	β	γ	ϵ
δ	α	β	β	δ	ϵ	ζ
ϵ	γ	γ	γ	ϵ	ϵ	ζ
ζ	δ	ϵ	ϵ	ζ	ζ	ζ

$\langle A, \star \rangle$

$*$	1	0	-1
1	1	1	0
0	1	0	-1
-1	0	-1	-1

$\langle B, * \rangle$

映射

$$\begin{aligned} f(\alpha) &= 1 & f(\beta) &= 1 & f(\gamma) &= 1 \\ f(\delta) &= 0 & f(\epsilon) &= 0 & f(\zeta) &= -1 \end{aligned}$$

明显地是由代数系统 $\langle A, \star \rangle$ 到 $\langle B, * \rangle$ 的一个同态映射。假如把代数系统 $\langle A, \star \rangle$ 看作是对六个带电粒子 $\alpha, \beta, \gamma, \delta, \epsilon, \zeta$ 相互作用的详尽描述。如果 α, β, γ 是带正电荷的粒子， δ, ϵ 是中性粒子， ζ 是带负电荷粒子，那么，我们就可用 1, 0, -1 分别表示这三类粒子，这就是映射 f 所具有的特性。若记

$$B = \{1, 0, -1\}$$

那么，代数系统 $\langle B, * \rangle$ 描述了三类粒子的相互作用，它正好是代数系统 $\langle A, \star \rangle$ 的粗糙描述。

Products and Quotients of Groups

积群和商群

Theorem 1

If

G_1 and G_2 are groups,

then

$$G = G_1 \times G_2$$

is a group with binary operation defined by

$$(a_1, b_1)(a_2, b_2) = (a_1a_2, b_1b_2).$$

Proof

By Theorem 1, Section 9.3, we have that G is a semigroup.

The existence of an identity and inverses is easy to verify.

Q.E.D.

证明课后练习

Example 1

Let G_1 and G_2 be the group Z_2 . For simplicity of notation, we shall write the elements of Z_2 as 0 and 1, respectively, instead of $\bar{0}$ and $\bar{1}$. Then the multiplication table of $G = G_1 \times G_2$ is given in Table.

$$G_1 = (Z_2, \oplus)$$

$$= \{0, 1\}$$

$$G_2 = (Z_2, \oplus)$$

$$= \{0, 1\}$$

$$G = G_1 \times G_2$$

Table 9.10 Multiplication Table of $Z_2 \times Z_2$

	$(\bar{0}, \bar{0})$	$(\bar{1}, \bar{0})$	$(\bar{0}, \bar{1})$	$(\bar{1}, \bar{1})$
$(\bar{0}, \bar{0})$	$(\bar{0}, \bar{0})$	$(\bar{1}, \bar{0})$	$(\bar{0}, \bar{1})$	$(\bar{1}, \bar{1})$
$(\bar{1}, \bar{0})$	$(\bar{1}, \bar{0})$	$(\bar{0}, \bar{0})$	$(\bar{1}, \bar{1})$	$(\bar{0}, \bar{1})$
$(\bar{0}, \bar{1})$	$(\bar{0}, \bar{1})$	$(\bar{1}, \bar{1})$	$(\bar{0}, \bar{0})$	$(\bar{1}, \bar{0})$
$(\bar{1}, \bar{1})$	$(\bar{1}, \bar{1})$	$(\bar{0}, \bar{1})$	$(\bar{1}, \bar{0})$	$(\bar{0}, \bar{0})$

Example 1

$|G|=4$, 所以 G 同构于 V (*Klein*) or Z_4

Since G is a group of order 4, it must be isomorphic to V or to Z_4 , the only groups of order 4.

The function

$f: V \rightarrow Z_2 \times Z_2$ defined by

$f(e) = (0, 0), f(a) = (1, 0),$

$f(b) = (0, 1),$ and $f(c) = (1, 1)$

is an isomorphism.

Table 9.5

	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	e	a
c	c	b	a	e

Table 9.7

	e	a	b	c
e	e	a	b	c
a	a	b	c	e
b	b	c	e	a
c	c	e	a	b

Table 9.6

	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	a	e
c	c	b	e	a

Table 9.8

	e	a	b	c
e	e	a	b	c
a	a	c	e	b
b	b	e	c	a
c	c	b	a	e

Note-推广

If we repeat Example 1 with Z_2 and Z_3 , we find that $Z_2 \times Z_3 \cong Z_6$.

It can be shown, in general, that

$$Z_m \times Z_n \cong Z_{mn}$$

if and only if

$\text{GCD}(m, n) = 1$. — m, n 互质时 $Z_m \times Z_n$ 同构 Z_{mn}

定理1推广: $G_1, G_2, G_3 \dots G_n$ 是群, 则 $G_1 \times G_2 \dots G_{n-1} \times G_n$ 也是群

例:

$$\mathbb{Z}_2 \times \mathbb{Z}_3 \cong \mathbb{Z}_6$$

$$\mathbb{Z}_2 \times \mathbb{Z}_3$$

生成元?

$\oplus$	(0,0)	(0,1)	(0,2)	(1,0)	(1,1)	(1,2)
(0,0)	(0,0)	(0,1)	(0,2)	(1,0)	(1,1)	(1,2)
(0,1)	(0,1)	(0,2)	(0,0)	(1,1)	(1,2)	(1,0)
(0,2)	(0,2)	(0,0)	(0,1)	(1,2)	(1,0)	(1,1)
(1,0)	(1,0)	(1,1)	(1,2)	(0,0)	(0,1)	(0,2)
(1,1)	(1,1)	(1,2)	(1,0)	(0,1)	(0,2)	(0,0)
(1,2)	(1,2)	(1,0)	(1,1)	(0,2)	(0,0)	(0,1)

定义同构
映射?

$\oplus$	[0]	[1]	[2]	[3]	[4]	[5]
[0]	[0]	[1]	[2]	[3]	[4]	[5]
[1]	[1]	[2]	[3]	[4]	[5]	[0]
[2]	[2]	[3]	[4]	[5]	[0]	[1]
[3]	[3]	[4]	[5]	[0]	[1]	[2]
[4]	[4]	[5]	[0]	[1]	[2]	[3]
[5]	[5]	[0]	[1]	[2]	[3]	[4]

Example 2

+	0	1
0	0	1
1	1	0

Let $B = \{0, 1\}$ be the group, where $+$ is defined as above

Then $B^n = B \times B \times \dots \times B$ (n factors) is a group with operation $\oplus$ defined by

$$\begin{aligned} (x_1, x_2, \dots, x_n) \oplus (y_1, y_2, \dots, y_n) \\ = (x_1 + y_1, x_2 + y_2, \dots, x_n + y_n). \end{aligned}$$

The identity of B^n is $(0, 0, \dots, 0)$, and every element is its own inverse.

单位元 $(0, 0, \dots, 0)$, 每个元素是自己的逆

Theorem 2-商群

Let R be a **congruence relation** on the group $(G, *)$.
Then the semigroup $(G/R, \odot)$ is a group, where the
operation $\odot$ is defined on G/R by
$$[a] \odot [b] = [a * b]$$

Proof: G 是群一定是独异点，由9.3节**推论1**知 G/R 为独异点，
只需再证明每个元素有逆。

Since a group is a monoid, G/R is a monoid.

Let $[a] \in G/R$.

Then $[a^{-1}] \in G/R$, and

$$[a] \odot [a^{-1}] = [a * a^{-1}] = [e].$$

$$\text{So } [a]^{-1} = [a^{-1}].$$

Hence $(G/R, \odot)$ is a group.

Corollary 1

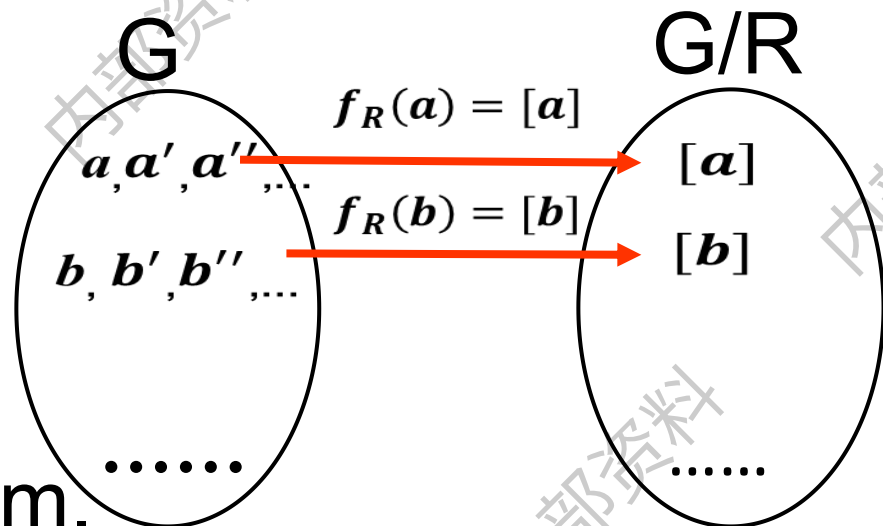
If R is a congruence relation on a group G ,
then the function

$$f_R: G \rightarrow G/R$$

given by

$$f_R(a) = [a]$$

is a group homomorphism.



R 是群 G 上的一个同余关系，那么由 $f_R(a) = [a]$
给出的函数 $f_R: G \rightarrow G/R$ 是一个同态

Corollary 2

-同态基本定理

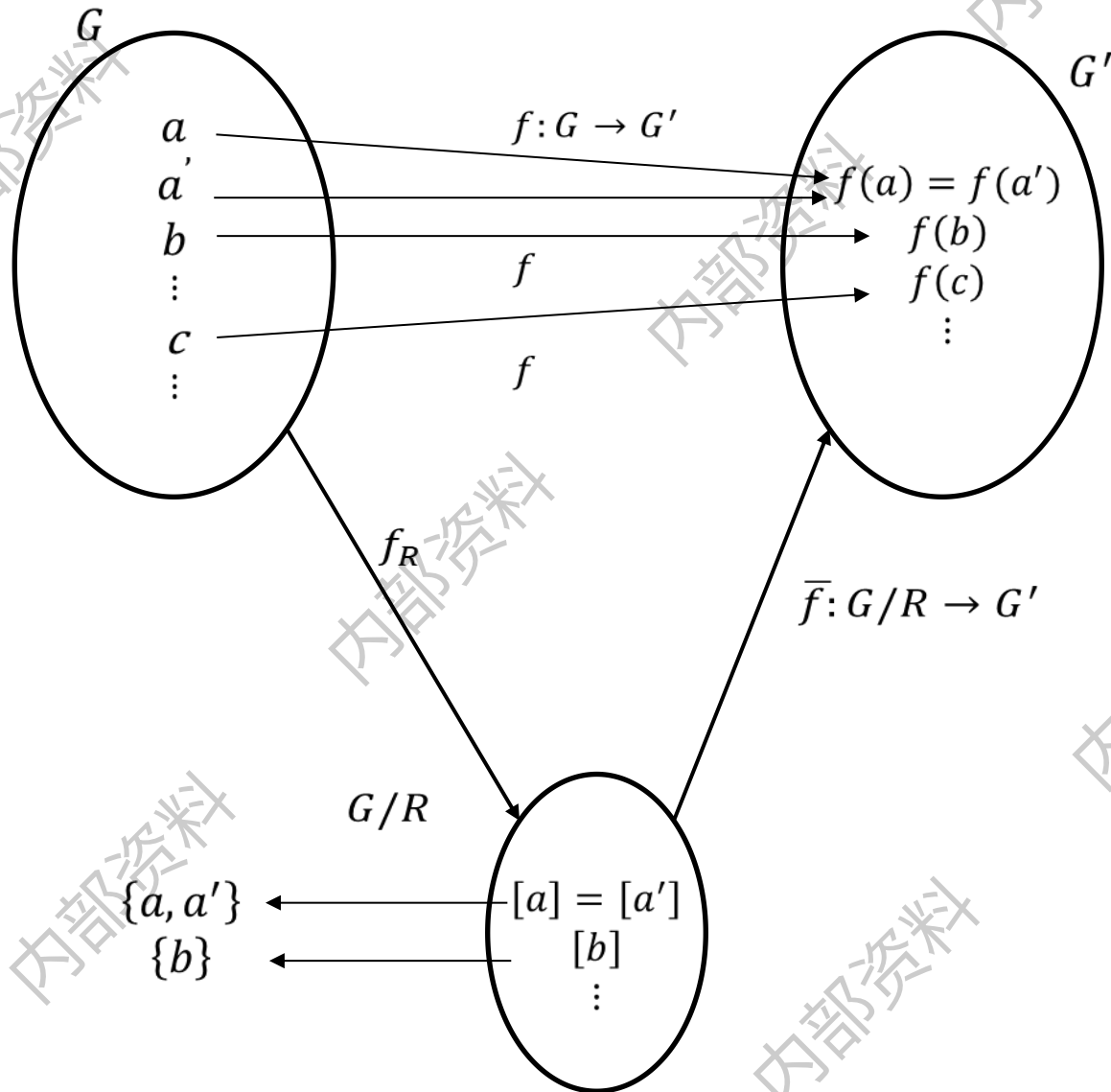
If $f: G \rightarrow G'$ is a homomorphism from the group $(G, *)$ **onto** the group $(G', *)$, and

R is the relation defined on G by

$a R b$ **if and only if** $f(a) = f(b)$, for a and b in G

Then

- R is a congruence relation.
- The function $\bar{f}: G/R \rightarrow G'$, given by $\bar{f}([a]) = f(a)$ is an isomorphism from the group $(G/R, \odot)$ onto the group $(G', *)$.



陪集

陪集 (coset) 定义及性质

Definition—陪集定义

Let

H be a subgroup of a group G

$$a \in G$$

The *left coset* (左陪集) of H in G determined by a is the set

$$aH = \{ah \mid h \in H\}$$

The *right coset* (右陪集) of H in G determined by a is the set

$$Ha = \{ha \mid h \in H\}$$

A subgroup H of G is *normal* (正规子群) if

$$aH = Ha, \text{ for all } a \text{ in } G$$

Definition—陪集定义

例 $(\mathbb{Z}_6, \oplus)$, 子群 $H = \{[0], [2], [4]\}$

$$[0] H = \{[0] \oplus [0], [0] \oplus [2], [0] \oplus [4]\} = \{[0], [2], [4]\}$$

$$[1] H = \{[1] \oplus [0], [1] \oplus [2], [1] \oplus [4]\} = \{[1], [3], [5]\}$$

$$[2] H = \{[2] \oplus [0], [2] \oplus [2], [2] \oplus [4]\} = \{[0], [2], [4]\}$$

$$[3] H = \{[3] \oplus [0], [3] \oplus [2], [3] \oplus [4]\} = \{[1], [3], [5]\}$$

$$[4] H = \{[4] \oplus [0], [4] \oplus [2], [4] \oplus [4]\} = \{[0], [2], [4]\}$$

$$[5] H = \{[5] \oplus [0], [5] \oplus [2], [5] \oplus [4]\} = \{[1], [3], [5]\}$$

$$[0] H = [2] H = [4] H$$

$$[1] H = [3] H = [5] H$$

任意两个左陪集，要么相等，要么不相交
什么情况相等，什么时候相交？

Warning

If $Ha = aH$, it does not follow that, for $h \in H$
and $a \in G$, $ha = ah$.

But $ha = ah'$, where h' is some element in H .

$$Ha = \{h_1 a, h_2 a, \dots, h_n a\}$$

$$aH = \{ah_1, ah_2, \dots, ah_n\}$$

注意:两个集合相等,并不是对应位置
上的元素相等!

Note

证明目标: 如果 $a \in H$, 则左陪集 $aH=H$

证明思路, 相互包含: $H \subseteq aH$, $aH \subseteq H$

If H is a subgroup of G , we shall need in some applications to compute all the left cosets of H in G .

First, suppose that $a \in H$.

Then $aH \subseteq H$, --since H is a subgroup of G ; (封闭性)

Moreover, if $h \in H$, then $h=ah'$, where $h'=a^{-1}h \in H$,
so that $H \subseteq aH$.

Thus, if $a \in H$, then $aH=H$.

陪集性质

子群 $H \subseteq G$, $\forall a, b \in G$

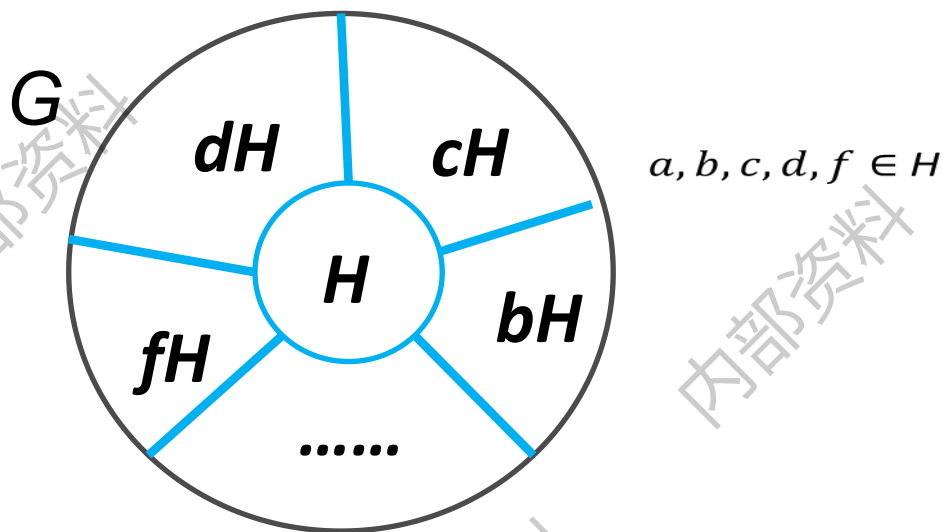
(1) $aH=H$, 当且仅当 $\forall a \in H$

(2) $aH=bH$, 当且仅当 $b \in aH$

(3) $aH \cap bH = \emptyset$, 当且仅当 $b \notin aH$

(4) $aH \cup bH \cup cH \dots = G$

(5) $\forall a, b \in G$, 如果 $a^{-1}b \in H$, 则 aRb , 为等价关系



思考：群 G 的正规子群是唯一的吗？

1. 所有阿贝尔群的子群都是正规子群
2. 所有平凡子群都是正规子群

陪集的个数？

拉格朗日定理： H 是有限群 G 的子群则 $|H|$ 可整除 $|G|$ 。

如何构造正规子群？

Example 3

Let

G be the symmetric group S_3 .

The subset $H = \{f_1, g_2\}$ is a subgroup of G .

Compute all the distinct left cosets of H in G .

H 的左陪集？

Example 3-1

Solution

$$H = \{f_1, g_2\}$$

If $a \in H$, then $aH = H$. Thus

$$f_1 H = g_2 H = H.$$

$$f_2 H = \{f_2, g_1\}$$

$$f_3 H = \{f_3, g_3\}$$

$$g_1 H = \{g_1, f_2\} = f_2 H$$

$$g_3 H = \{g_3, f_3\} = f_3 H$$

The distinct left cosets of H in G are

H , $f_2 H$, and $f_3 H$.

Example 3-2

$H = \{f_1, g_2\}$, 求右陪集?

Solution

If $a \in H$, then $Ha = H$. Thus

$$Hf_1 = Hg_2 = H = \{f_1, g_2\}.$$

$$Hf_2 = \{f_2, g_3\}$$

$$Hf_3 = \{f_3, g_1\}$$

$$Hg_1 = \{g_1, f_3\} = Hf_3$$

$$Hg_3 = \{g_3, f_2\} = Hf_2$$

有些左右陪集不相等!

$$f_2H = \{f_2, g_1\}$$

$$Hf_2 = \{f_2, g_3\}$$

Example 3-3

$H = \{f_1, f_2, f_3\}$, 求左、右陪集?

Solution

$$g_1 H = H g_1 = \{g_1, g_2, g_3\}.$$

$$f_1 H = H f_1 = \{f_1, f_2, f_3\}$$

左陪集=右陪集

$$aH = H a$$

$$ah = ha ?$$

Theorem 3

Let

R be a congruence relation on a group G
 $H = [e]$, the equivalence class containing the identity.

Then

- $[a] = aH = Ha$, for each $a \in G$
- H is a normal subgroup of G

Proof(1/2)

Let

a and b be any elements in G

Then $([a] = aH?)$

$b \in [a]$ *if and only* if $[b] = [a]$.

if and only if $[e] = [a]^{-1}[a] = [a]^{-1}[b] = [a^{-1}b]$.

G/R is a group

$$\begin{cases} H = [e] \\ [e] = [a^{-1}b] \end{cases} \Rightarrow H = [e] = [a^{-1}b]$$

说明 $a^{-1}b$ 是 H 的代表元,

所以 $a^{-1}b \in H$ or $aa^{-1}b \in aH$. (两边同左乘 a) $\Leftrightarrow b \in aH$

($\because \forall b \in [a] \Rightarrow b \in aH, \therefore [a] \subseteq aH$.)

同理可证 $aH \subseteq [a]$, 所以 $[a] = aH$

R等价关系, a , b 都可以作为代表元表示同一个等价类

G/R 商群,
 $[a][b][e]$ 都是其中的元素,
所以单位元 $[e] = [a]^{-1}[a]$

Proof(2/2)

Similarly,

$$b \in [a] \text{ if and only if } H = [e] = [b][a]^{-1} = [ba^{-1}].$$

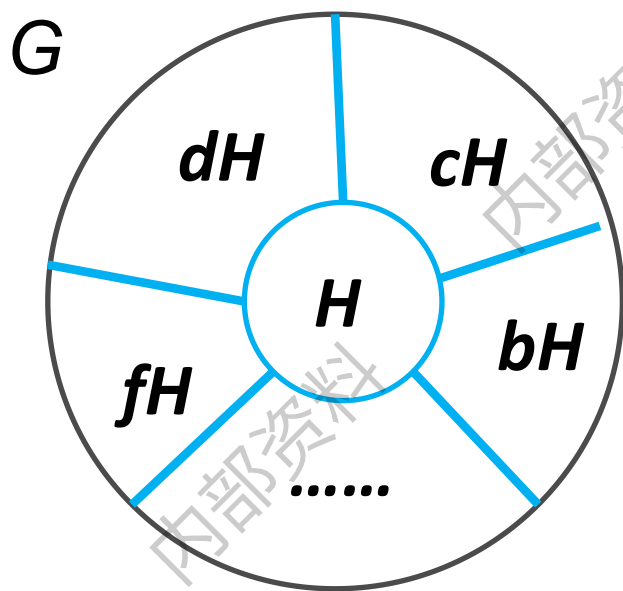
$$[a] = Ha.$$

Thus $[a] = aH = Ha$, and H is normal.

Q.E.D.

回顾陪集

R 同余关系对 G 形成划分，每一个块都是一个等价类，其中包含单位元 e 的等价类 H 。



例 $(\mathbb{Z}_6, \oplus)$, 子群

$H = \{[0], [2], [4]\}$

$[0] \ H = \{[0], [2], [4]\}$

$[1] \ H = \{[1], [3], [5]\} = [1]$

$[3] \ H = \{[1], [3], [5]\} = [3]$

$[5] \ H = \{[1], [3], [5]\} = [5]$

$[a] = aH = Ha$

Note

Combining Theorem 3 with Corollary 1, we see that in this case the quotient group G/R consists of all the left cosets of $N = [e]$. The operation in G/R is given by

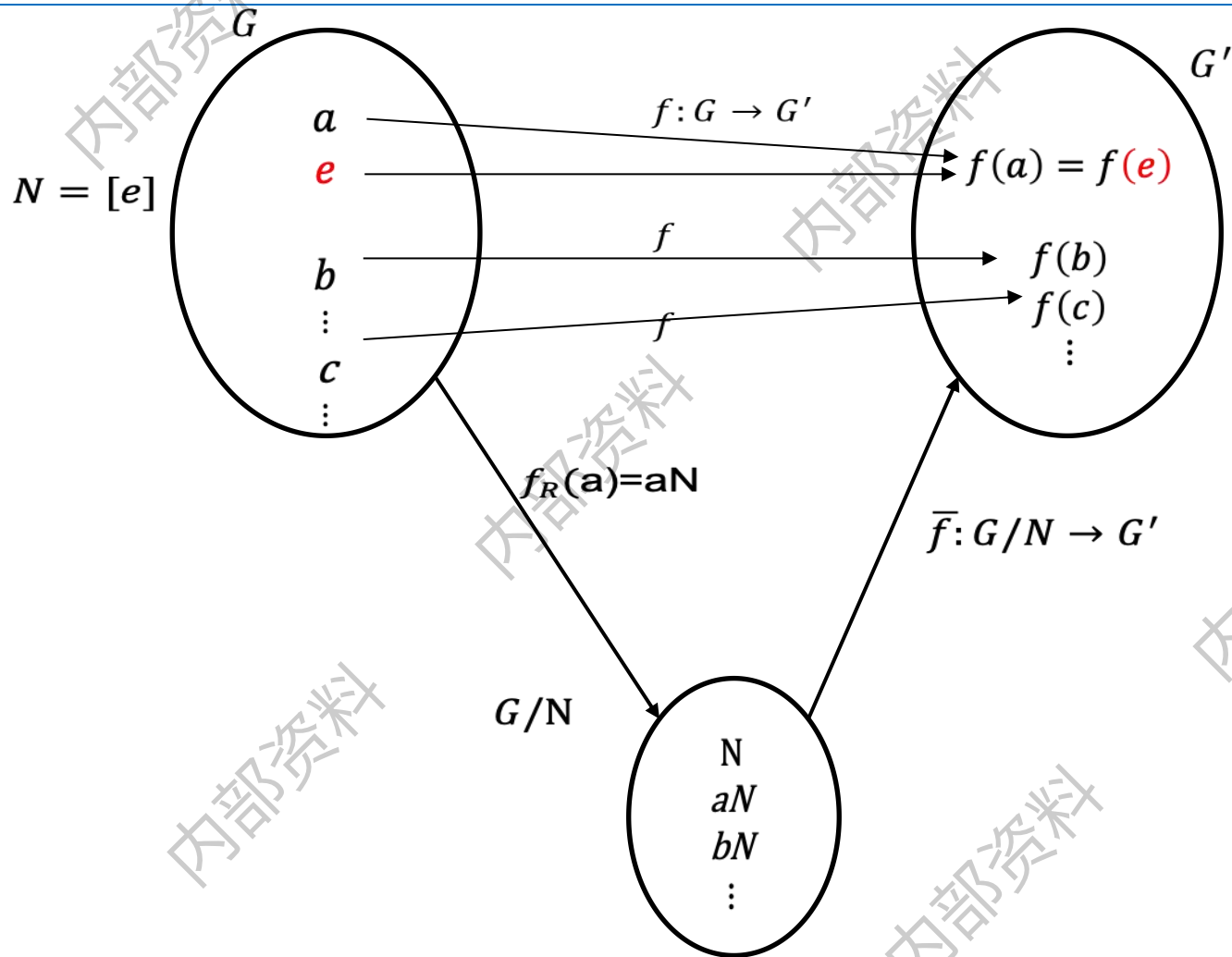
$$(aN)(bN) = [a] \odot [b] = [ab] = abN \text{ (这里的 } N=[e] \text{ 就是正规子群 } H \text{ 的一般标记, } [a]=aN, [b]=bN)$$

and the function $f_R: G \rightarrow G/R$, defined by

$$f_R(a) = aN$$

is a homomorphism from G onto G/R . For this reason, we will often write G/R as G/N .

$$f_R(a) = aN \text{ (根据theorem3, } aN=[a])$$



Theorem 4

Let

N be a normal subgroup of a group G

R be the following relation on G , aRb *if and only if*
 $a^{-1}b \in N$. (N 是正规子群, 如果 $a^{-1}b \in N$, 则
 aRb)

前提

Then

(a) R is a congruence relation on G .

(b) N is the equivalence class $[e]$ relative to R , where
 e is the identity of G .

1. R 同余
2. $N=[e]$

结论

Proof (a)

R is an equivalence relation (证等价)

Let $a \in G$.

$a R a$, since $a^{-1}a = e \in N$,
 R is reflexive.

(自反)

Suppose that $a R b$

$a^{-1}b \in N$.

$(a^{-1}b)^{-1} = b^{-1}a \in N$,

$b R a$.

R is symmetric.

(对称)

Suppose that $a R b$ and $b R c$.

$a^{-1}b \in N$ and $b^{-1}c \in N$.

$(a^{-1}b)(b^{-1}c) = a^{-1}c \in N$,

$a R c$.

R is transitive.

(传递)

Proof (a)

$a R b$ and $c R d$

Imply

$(a * c) R (b * d)$

R is a congruence relation on G . (证同余)

Suppose that $a R b$ and $c R d$.

$(ac)^{-1} bd \in N?$

Then $a^{-1}b \in N$ and $c^{-1}d \in N$

Since N is normal, $Nd = dN$, 对任意 $n_1 \in N$, 存在 $n_2 \in N$, 使得 $n_1 d = d n_2$

因为 $a^{-1}b \in N$, 所以 $a^{-1}bd = d n_2$

$$(ac)^{-1}bd = (c^{-1}a^{-1})(bd) = c^{-1}(\underbrace{a^{-1}b}_{n_1})d = (c^{-1}d)n_2 \in N$$

$$(ac)^{-1}bd \in N \Rightarrow ac R bd.$$

Hence R is a congruence relation on G .

Proof (b)

证明: $N=[e]$?

Suppose that $x \in N$.

$$x^{-1} = x^{-1}e \in N \quad (\text{因为 } N \text{ 为子群, } e \in N)$$

$$x R e$$

$$x \in [e]$$

$$N \subseteq [e]$$

Conversely, if $x \in [e]$,

$$x R e$$

$$x^{-1}e = x^{-1} \in N \quad \text{--- } a R b \text{ if and only if } a^{-1}b \in N.$$

$$x \in N$$

$$[e] \subseteq N$$

Hence $N = [e]$.

Note—三种等价表述

Following three are equivalent

a congruence relation R on G

a normal subgroup H of G

a homomorphism from G to G/R or G'

小节

1. 群 G 上的同余关系的等价类，总是某个正规子群的陪集
2. 群 G 的某个正规子群的陪集恰好是关于 G 上某个同余关系的等价类

根据定义求陪集，必须要很多次计算 aH ；

利用同余关系 R 求陪集，实际就是计算等价类，变得相对方便

Corollary 2

Let

f be a homomorphism from a group $(G, *)$ **onto**
a group $(G', *)$

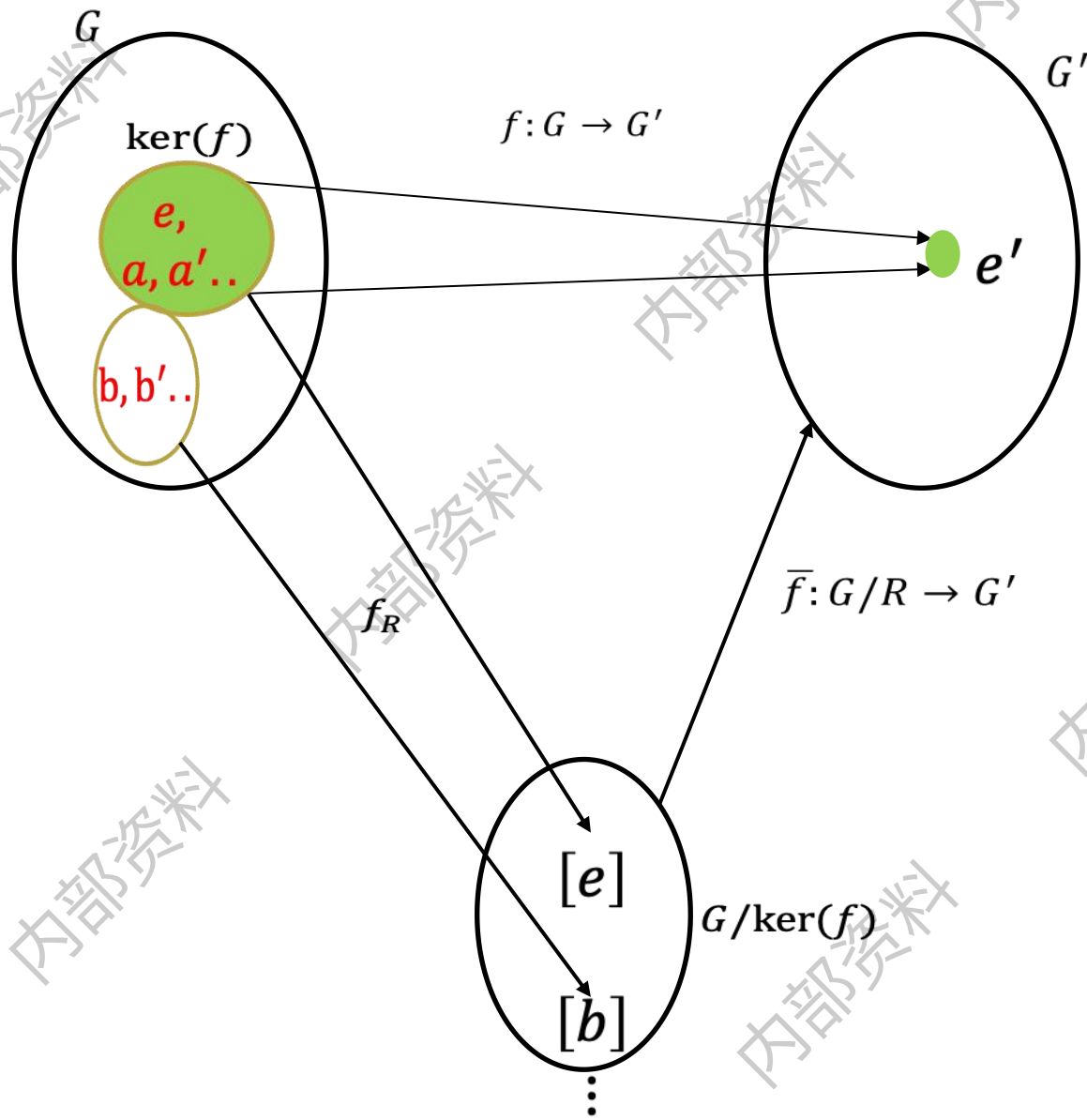
the **kernel**(核) of f , $\ker(f)$, be defined by

$$\ker(f) = \{a \in G \mid f(a) = e'\}.$$

Then

$\ker(f)$ is a normal subgroup of G .

The quotient group $G/\ker(f)$ is isomorphic to G' .



Example 6

Consider the homomorphism f from $\mathbb{Z}$ onto $\mathbb{Z}_n$ defined by $f(m) = [r]$, where r is the remainder when m is divided by n . Find $\ker(f)$.

Solution

An integer m in $\mathbb{Z}$ belongs to $\ker(f)$ if and only if $f(m) = [0]$

if and only if m is a multiple of n

Hence $\ker(f) = n\mathbb{Z}$.

Example

$$B = \{0, 1\}$$

$$B^5 = \{ \begin{array}{llll} 00000 & 01011 & 10110 & 11101 \\ 01000 & 00011 & 11110 & 10101 \\ 10000 & 11011 & 00110 & 01101 \\ 11000 & 10011 & 01110 & 00001 \\ 00001 & 01010 & 10111 & 11100 \\ 00010 & 01001 & 10100 & 11111 \\ 10001 & 11010 & 00111 & 01100 \\ 10010 & 11001 & 00100 & 01111 \end{array} \}$$

$$B^3 = \{ \begin{array}{l} 000 \\ 011 \\ 110 \\ 101 \\ 001 \\ 010 \\ 111 \\ 100 \end{array} \}$$

$$f = x * H$$

$$\text{Ker}(f) = \{00000, 01011, 10100, 11101\}$$

$$H = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

正规子群小结

1. 满足 $aH=Ha$, H 为正规子群
2. Abel 群的任何子群是正规子群
3. 群 G 中的同余关系 R , 确定的等价类 $N=[e]$
4. 同态核, $\text{Ker}(f)$

正规子群的判别（补）

正规子群 $H \subseteq G$ ，判别定理：

1. $aH = Ha, \forall a \in G$

2. $\forall a \in G, aHa^{-1} = H$

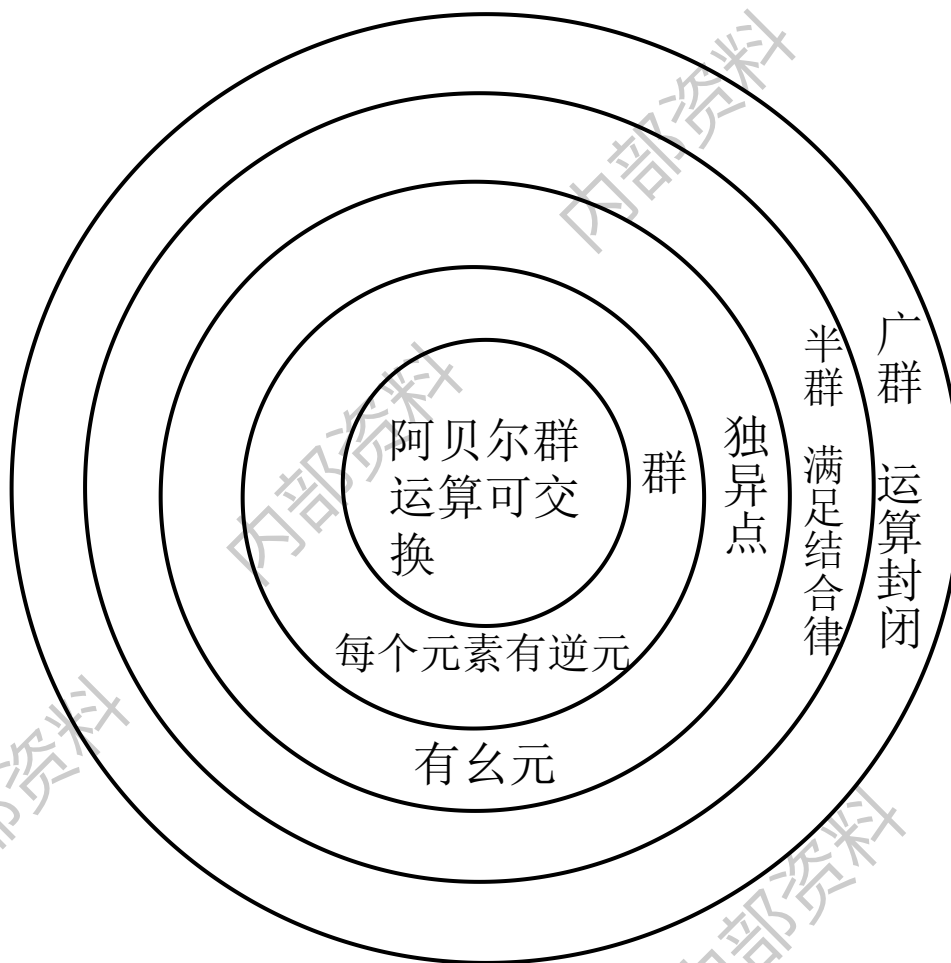
3. $\forall a \in G, h \in H, aha^{-1} \in H$

三个判别条件等价

Homework

4, 22, 34@9.5

$(S, *)$



内部资料

内部资料

群

有限群一阶、二阶、三阶、四阶 (V & Z_4)

置换群 S_n

循环群

$$(S \times T, *) (s_1, t_1) * (s_2, t_2) = (s_1 * s_2, t_1 * t_2)$$

乘积(半)群

同余关系 R, S 上的等价关系 $a R b, c R d \Rightarrow a * c R b * d$

产生

商(半)群

$$(S/R, @) [a]@[b] = [a * b]$$

H 为 S 的子集

本身为(半)群

子(半)群

正规子群: G 中所有的 a $aH = Ha$

多(半)群

结构之间

同构

S 到 T 存在一一映射(双射)

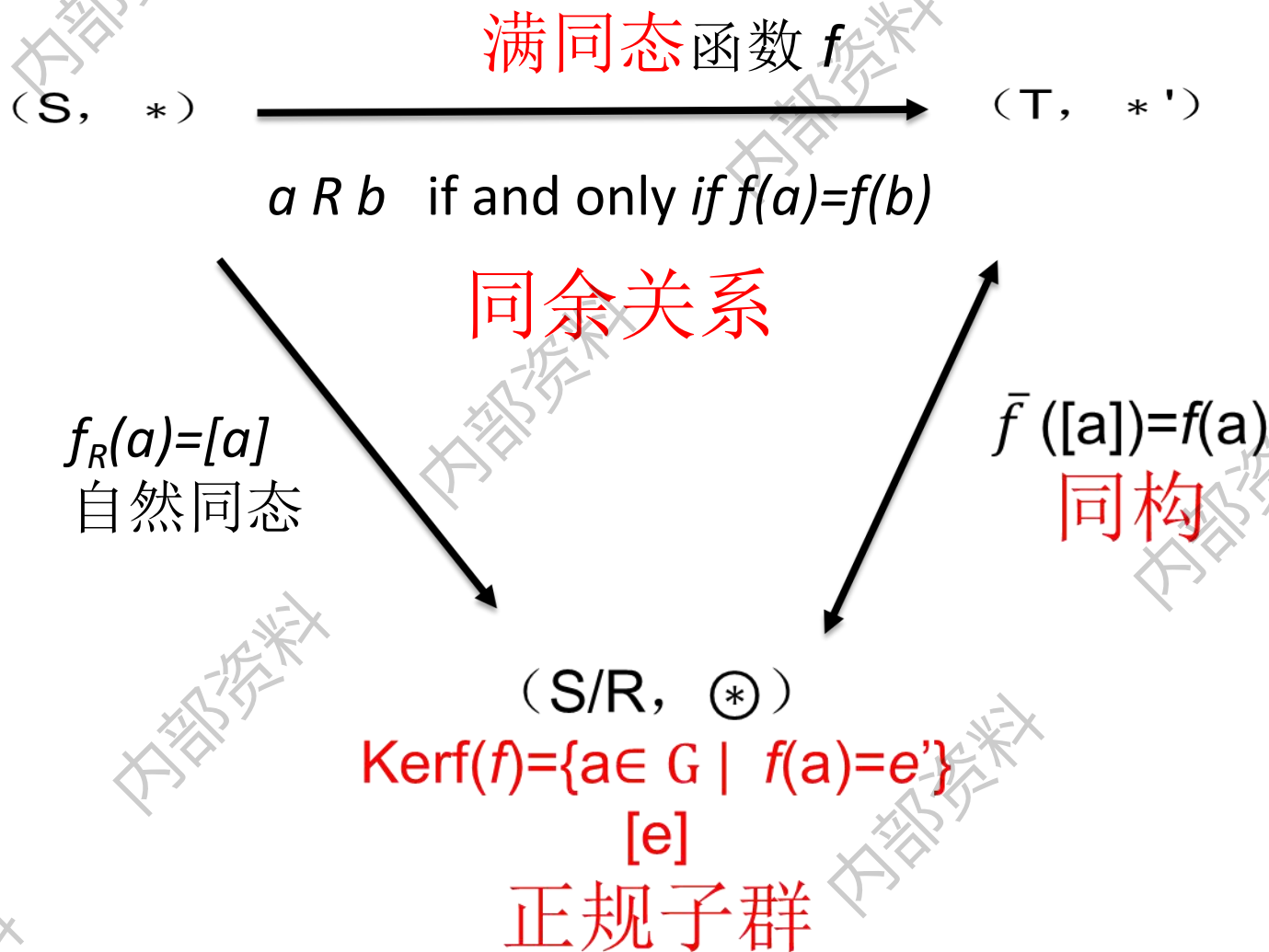
$$f(a * b) = f(a) *' f(b)$$

$(s, *) \rightarrow (t, *)$ S 到 T 存在全函数

同构

$$f(a * b) = f(a) *' f(b)$$

同态基本定理



$\leq$

Thanks for your attention!

Coming up next ...

Groups and Coding